

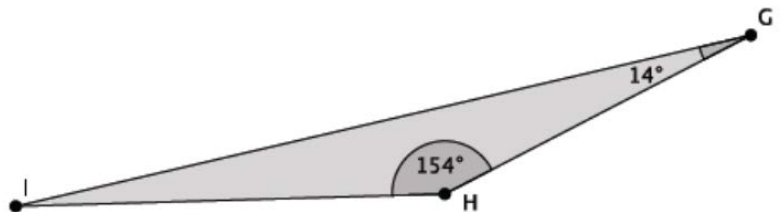
Grade 8
Unit 13
Lesson 3 Practice

Name Answer Key
Date _____
Period _____

1. The Triangle Sum Theorem states that the ☐ sum
☒ difference of the measures of the ☐ exterior
☒ interior
☐ adjacent angles of a triangle is 180° .

2. Find the measure of Angle HIG .

$$180 - 154 - 14 = 12^\circ$$



Use Triangle BAG for questions 3–5.

3. Write an equation that can be used to find the value of y .

$$y + (2y - 30) + (y + 10) = 180$$

4. Solve the equation.

$$y + (2y - 30) + (y + 10) = 180$$

$$4y - 20 = 180$$

$$4y = 200$$

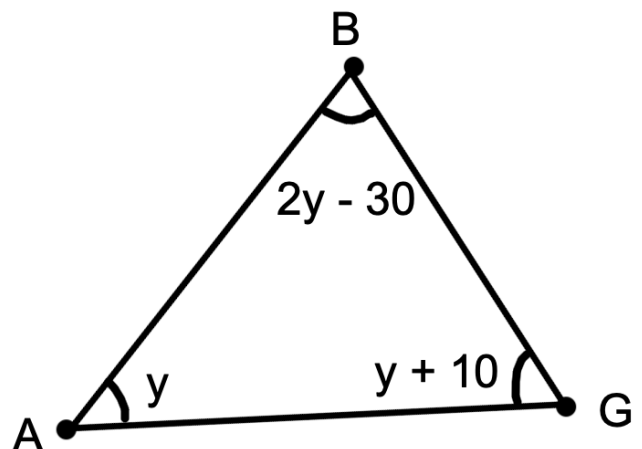
$$y = 50$$

5. Find the measures of all the angles of Triangle BAG .

$$\angle BAG = 50^\circ$$

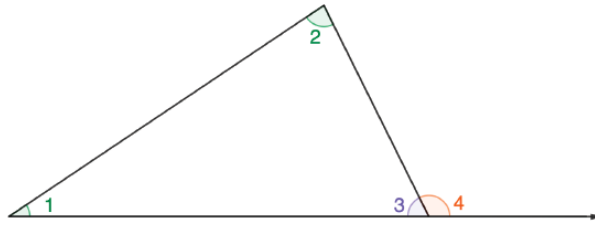
$$\angle AGB = 50 + 10 = 60^\circ$$

$$\angle GBA = 2(50) - 30 = 70^\circ$$



6. Which statement is *not true* about the triangle?

- A. $\angle 1 + \angle 2 = \angle 4$
- B. $\angle 1 + \angle 2 + \angle 3 = 180^\circ$
- C. $\angle 1 + \angle 3 = \angle 180^\circ$**
- D. $\angle 3 + \angle 4 = \angle 180^\circ$



7. Every interior angle of a triangle has 2 possible exterior angles. The exterior angles have the same measure since they are

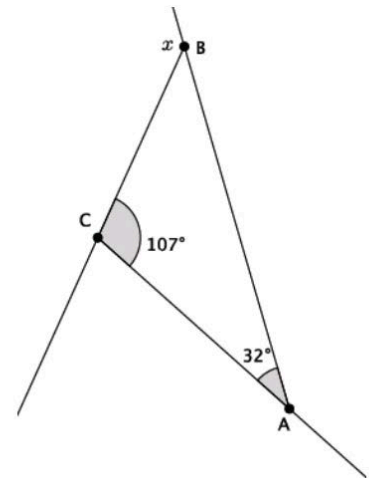
- ☐ complementary.
- ☐ supplementary.
- ☒ vertical.**

8. Consider the diagram shown. Find the value of x .

$$180 - 107 - 32 = 41^\circ$$

$$x = 180 - 41$$

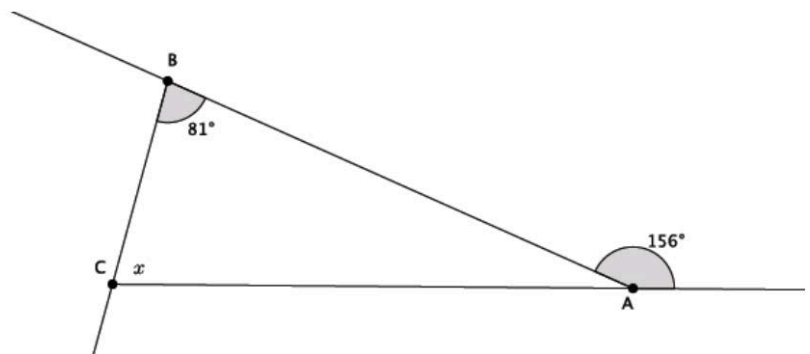
$$x = 139^\circ$$



Use the figure shown for questions 9–10.

9. Write an equation to find the value of $\angle BCA$.

$$x + 81 = 156$$



10. Solve the equation.

$$x + 81 = 156$$

$$x = 75^\circ$$